



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

SCORE - II

14-05-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Main)

TEST DATE : 14-05-2021

ANSWER KEY

PHYSICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	B	C	D	B	A	A	D	B
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	B	A	D	A	C	A	B
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	30.00	0.67	2.82	2.00	1.12	0.34 to 0.35	8.00	3.21 to 3.24	0.50	2.00

CHEMISTRY

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	D	D	D	C	D	C	A
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	D	B	D	A	C	D	C	C
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	60.00	117.00	5.00	5.00	56.00	4.00	4.00	-1656.51	4.00	6.00

MATHEMATICS

SEC-A	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	C	A	B	B	C	D	D
SEC-A	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	C	C	B	C	C	D	B	A
SEC-B	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.50	1.00	3.00	40.00	36.00	0.00	1.00	51.80	37.50	3.20

HINT – SHEET

PHYSICS

SEC-A

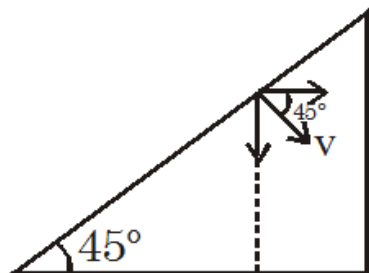
1. Ans (A)

Let the ball hit the inclined with velocity 'V'

$$V \cos 45^\circ = 10 \cos 60^\circ$$

$$V = \frac{10}{2} \times \sqrt{2} = 5\sqrt{2} \text{ m/s}$$

$$v_y = u_y + a_y t [V_y = V \sin 45^\circ = 5 \text{ m/s}]$$



$$-5 = 10 \sin 60^\circ - gt$$

$$-5 - 5\sqrt{3} = -10t$$

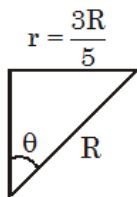
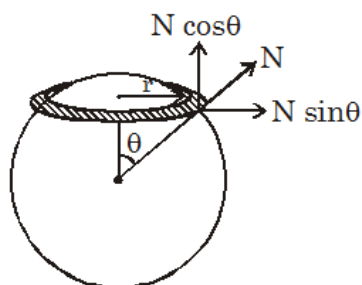
$$t = \frac{5(\sqrt{3} + 1)}{10}; t = \frac{\sqrt{3} + 1}{2}$$

$$x = u_x \times t$$

$$= 10 \cos 60^\circ \times \frac{\sqrt{3} + 1}{2}$$

$$x = \frac{5(\sqrt{3} + 1)}{2}$$

2. Ans (C)



$$\sin \theta = \frac{3R}{5R}$$

$$\sin \theta = \frac{3}{5}$$

$$\theta = 37^\circ$$

$$N \sin \theta = T (2\pi) \dots (I)$$

$$N \cos \theta = W \dots (II)$$

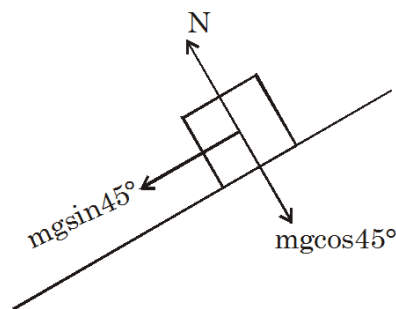
$$\tan \theta = \frac{T(2\pi)}{W}$$

$$\frac{3}{4} = \frac{T(2\pi)}{W}$$

$$T = \frac{3W}{8\pi}$$

3. Ans (B)

F.B.D. of block (m)

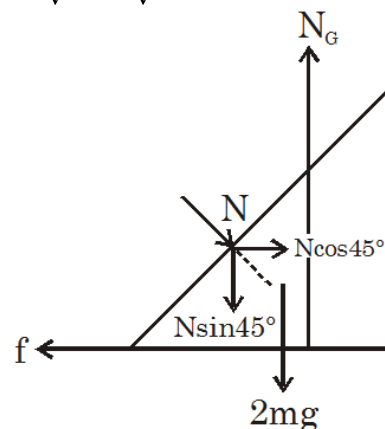


$$N = mg \cos 45^\circ = \frac{mg}{\sqrt{2}}$$

F.B.D. of wedge

$$N_G = N \sin 45^\circ + 2mg$$

$$= \frac{mg}{\sqrt{2}} \times \frac{1}{\sqrt{2}} + 2mg$$



$$= \frac{5}{2} mg$$

$$f_{\text{required}} \leq \mu N_G$$

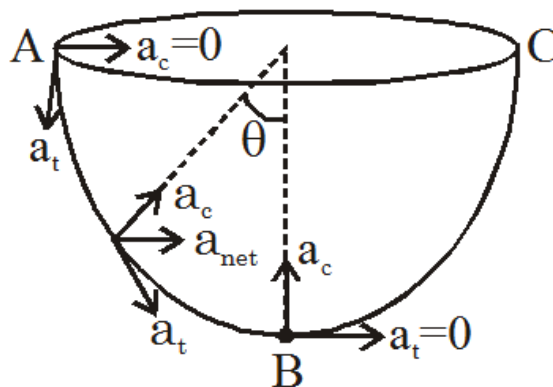
$$N \cos 45^\circ \leq \mu \left(\frac{5}{2} mg \right)$$

$$\frac{1}{2} \leq \mu \left(\frac{5}{2} \right)$$

$$\mu \geq \frac{1}{5}$$

$$\mu_{\text{minimum}} = 0.2$$

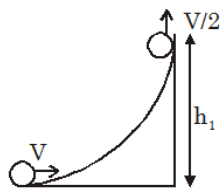
4. Ans (C)



5. **Ans (D)**

Velocity of ball at bottom of the track = 'v'
Applying W.E. theorem

$$mgh = \frac{1}{2}mv^2 - 0$$



$$v = \sqrt{2gh}$$

Maximum velocity of the ball is at bottom point.

Applying W.E.T.

$$-mgh_1 = \frac{1}{2}m\left(\frac{v}{2}\right)^2 - \frac{1}{2}mv^2$$

$$= \frac{1}{2}\left(\frac{v^2}{4} - v^2\right)$$

$$-gh_1 = -\frac{3v^2}{8}$$

$$gh_1 = \frac{3}{8}(2gh)$$

$$h_1 = \frac{3}{4}h$$

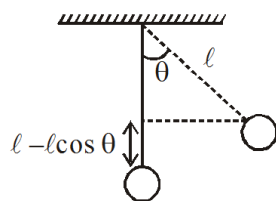
6. **Ans (B)**

At maximum speed, centripetal acceleration will also be maximum.

By W.E.T.

$$W_g + W_T + W_F = \Delta K$$

$$-mg\ell(1 - \cos\theta) + 0 + mg\ell\sin\theta = \frac{1}{2}mv^2 - 0$$



$$v^2 = -2g\ell + 2g\ell\cos\theta + 2g\ell\sin\theta$$

$$v = \sqrt{-2g\ell + 2g\ell\cos\theta + 2g\ell\sin\theta}$$

$$\text{for maximum velocity } \frac{dv}{d\theta} = 0$$

$$\frac{dv}{d\theta} = \frac{1}{2\sqrt{-2g\ell + 2g\ell\cos\theta + 2g\ell\sin\theta}}$$

$$\times (-2g\ell\sin\theta + 2g\ell\cos\theta)$$

$$\frac{dv}{d\theta} = 0$$

$$-2g\ell\sin\theta + 2g\ell\cos\theta = 0$$

$$2g\ell\sin\theta = 2g\ell\cos\theta$$

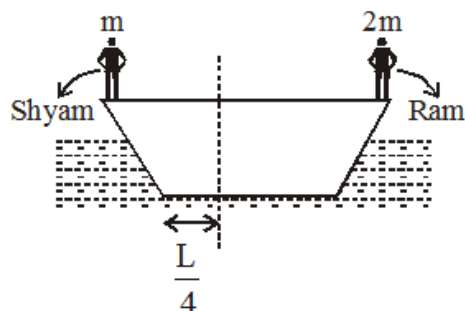
$$\sin\theta = \cos\theta$$

$$\theta = \frac{\pi}{4}$$

7. **Ans (A)**

Let mass of Shyam = m

$\therefore$ mass of Ram = 2 m



$$2m\left(L - \frac{L}{4}\right) + M\left(-\frac{L}{4}\right) + m\left(-\frac{L}{4} - L\right) = 0$$

$$\frac{6mL}{4} - \frac{ML}{4} - \frac{5mL}{4} = 0$$

$$\frac{mL}{4} = \frac{ML}{4}$$

$$\frac{M}{m} = 1$$

8. **Ans (A)**

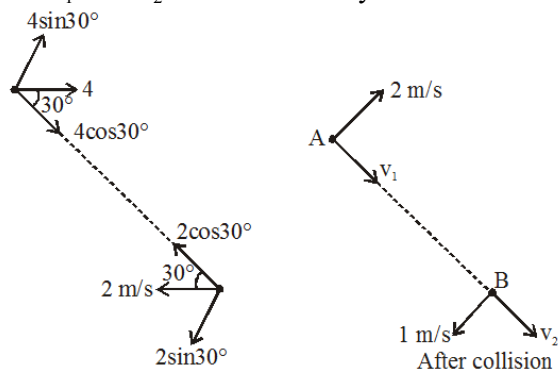
$$b\sin\theta = \pm\lambda$$

$$\sin\theta = \pm\frac{\lambda}{b} = \pm 2.25 \times 10^{-3}$$

$$\theta = \pm 2.25 \times 10^{-3}$$

$$\text{so angular divergence} = 2\theta = 4.5 \times 10^{-3} \text{ rad}$$

9. **Ans (D)**
Let V_1 and V_2 are final velocity after collision



Before collision

By conservation of momentum along L.O.C.

$$2 \times 2\sqrt{3} - 4\sqrt{3} = 2v_1 + 4v_2$$

$$v_1 + 2v_2 = 0$$

$$v_1 = -2v_2 \quad \dots(i)$$

velocity of separation along L.O.C.

$$e = \frac{\text{velocity of separation along L.O.C.}}{\text{velocity of approach along L.O.C.}}$$

$$\frac{1}{3} = \frac{v_2 - v_1}{2\sqrt{3} + \sqrt{3}}$$

$$v_2 - v_1 = \sqrt{3} \quad \dots(ii)$$

From equation (i) and (ii)

$$v_2 + 2v_2 = \sqrt{3}$$

$$v_2 = \frac{1}{\sqrt{3}}$$

$$v_1 = -\frac{2}{\sqrt{3}} \text{ m/s}$$

$$\vec{I} = \Delta P$$

$$|\vec{I}| = 4 \times \frac{1}{\sqrt{3}} - 4(-\sqrt{3})$$

$$|\vec{I}| = \frac{4}{\sqrt{3}} + 4\sqrt{3}$$

$$|\vec{I}| = \frac{16}{\sqrt{3}} \text{ N-s}$$

10. **Ans (B)**
 πR^2 Area $\rightarrow$ Mass M

$$\text{unit Area} \rightarrow \frac{M}{\pi R^2}$$

$$\frac{\pi R^2}{16} \text{ unit Area} \rightarrow \frac{M}{\pi R^2} \times \frac{\pi R^2}{16} = \frac{M}{16}$$

$$\text{Mass of small disc} = \frac{M}{16}$$

(M.O.I.)_{AB}

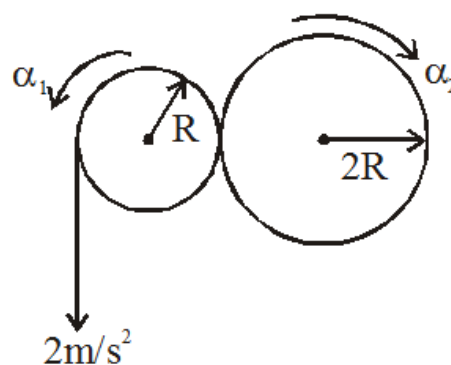
$$= \frac{MR^2}{4} - 2 \left(\frac{M}{16} \left(\frac{R}{4} \right)^2 \right) - 2 \left(\frac{M}{16} \left(\frac{R}{4} \right)^2 + \frac{M}{16} \left(\frac{3R}{4} \right)^2 \right)$$

$$= \frac{MR^2}{4} - \frac{2MR^2}{1024} - \frac{2MR^2}{1024} - 2 \left(\frac{9MR^2}{256} \right)$$

$$45 MR^2$$

$$\text{HS-4/16} = \frac{45 MR^2}{256}$$

11. **Ans (A)**



$$\alpha_1 R = \alpha_2 (2R) \quad \dots(i)$$

$$\alpha_1 R = 2 \quad \dots(ii)$$

$$\text{Also, } f(2R) = \frac{2 \times (2R)^2}{2} \alpha_2$$

$$\therefore f = 2N.$$

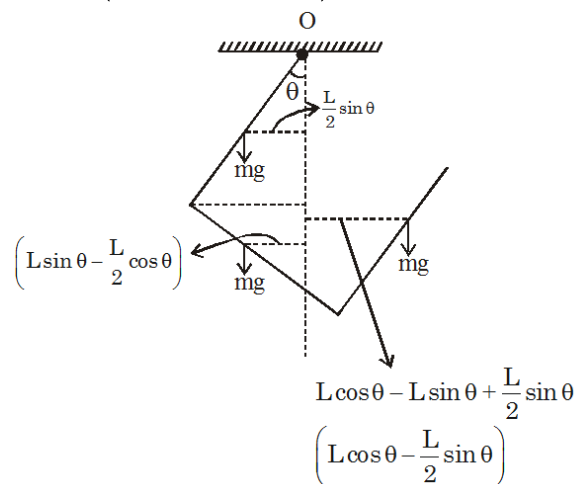
12. **Ans (C)**

Mass of each segment = m

net torque about point 'O' equals to zero

$$mg \frac{L}{2} \sin \theta + mg \left(L \sin \theta - \frac{L}{2} \cos \theta \right)$$

$$= mg \left(L \cos \theta - \frac{L}{2} \sin \theta \right)$$



$$\frac{L}{2} \sin \theta + L \sin \theta - \frac{L}{2} \cos \theta = L \cos \theta - \frac{L}{2} \sin \theta$$

$$2L \sin \theta = \frac{3L}{2} \cos \theta$$

$$\tan \theta = \frac{3}{4}$$

$$\theta = 37^\circ$$

13. Ans (B)

electric field $\vec{E} = 5\hat{j} \cos(3x + 4z)$

phase angle is independent of time

phase angle at $t = 0$, $3x + 4z$.

So phase angle for $\vec{B}$ will also be $3x + 4z$

because $\vec{E}$ & $\vec{B}$ oscillate in same phase.

direction of wave propagation

$$= \frac{3\hat{i} + 4\hat{k}}{\sqrt{3^2 + 4^2}} = \frac{3\hat{i} + 4\hat{k}}{5}$$

Let magnetic field vector $\vec{B} = a\hat{i} + b\hat{j} + d\hat{k}$

direction of wave propagation

$$= \frac{\vec{E} \times \vec{B}}{|\vec{E}| |\vec{B}|} = \frac{5\hat{j} \times (a\hat{i} + b\hat{j} + d\hat{k})}{5 \times (a^2 + b^2 + d^2)^{\frac{1}{2}}}$$

$$= \frac{-5a\hat{k} + 5d\hat{i}}{5(a^2 + b^2 + d^2)^{\frac{1}{2}}}$$

$$B = \frac{E}{c} = \frac{5}{c}$$

$$B = \sqrt{a^2 + b^2 + d^2} = \frac{5}{c}$$

$$\frac{3\hat{i} + 4\hat{k}}{5} = \frac{c \times 5}{5 \times 5} (d\hat{i} - a\hat{k})$$

$$3\hat{i} + 4\hat{k} = c (d\hat{i} - a\hat{k})$$

$$d = \frac{3}{c}, a = -\frac{4}{c}$$

$$\vec{B} = \frac{3}{c}\hat{k} - \frac{4}{c}\hat{i} = \frac{1}{c} (3\hat{k} - 4\hat{i})$$

so general equation of $\vec{B}$

$$\vec{B} = \frac{1}{c} (3\hat{k} - 4\hat{i}) \cos (3x + 4z - 5ct)$$

14. Ans (B)

Kinetic energy = K

$$-mgh = \Delta K$$

$$-mgh = \frac{K}{2} - K$$

$$mgh = \frac{K}{2}$$

$$2mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$4mgh = mv^2 + I \frac{v^2}{R^2} \quad \{\text{Let } I = n(mr^2)\}$$

$$4mgh = mv^2 + nmR^2 \frac{v^2}{R^2}$$

$$4mgh = mv^2 + nmv^2$$

$$40 \times \frac{15}{4} = v^2 (1 + n)$$

$$15 \times 10 = 100(1 + n)$$

$$(n + 1) = \frac{15}{10} = \frac{3}{2}$$

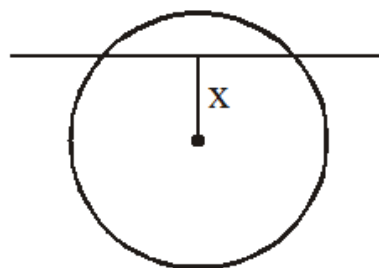
$$n = \frac{1}{2}$$

$$\left(I = \frac{mR^2}{2} \right)$$

15. Ans (A)

$$T = 2\pi \sqrt{\frac{I}{mgx}}$$

$$T = 2\pi \sqrt{\frac{\frac{mr^2}{4} + mx^2}{mgx}}$$



$$T = 2\pi \sqrt{\frac{r^2}{4gx} + \frac{x}{g}}$$

For T to be minimum

$$\frac{dT}{dx} = 0$$

$$-\frac{r^2}{4x^2} + 1 = 0$$

$$r^2 = 4x^2 \Rightarrow x = r/2$$

16. **Ans (D)**

$$A\omega = v_{\max} = 3.14$$

$$A.2\pi v = 3.14 \Rightarrow v = 5 \text{ Hz.}$$

On solving $A = 0.0001 \text{ m.}$

17. **Ans (A)**

After 5 sec, speed of detector = $50 - 10 \times 5 = 0$
and that of source = $0 + 10 \times 5 = 50 \text{ m/s}$ and
the source has fallen a distance

$$= \frac{1}{2} \times 10 \times 5^2 = 125 \text{ m}$$

and the detector has risen a distance

$$= 50 \times 5 - \frac{1}{2} \times 10 \times 5^2 = 125 \text{ m.}$$

$$\text{So } f' = 130 \left(\frac{300 - 0}{300 - 50} \right) = 156 \text{ Hz.}$$

18. **Ans (C)**

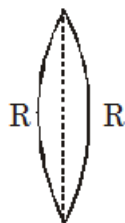
In resonance column experiment

$$\frac{\lambda}{4} = \ell_1 + e \text{ so,}$$

$$\ell_1 = \frac{\lambda}{4} - e = \frac{v}{4f} - 0.3d$$

$$= \frac{336 \times 100 \text{ cm}}{4 \times 1050} - (0.3) \left(\frac{20}{3} \text{ cm} \right) = 8 - 2 = 6 \text{ cm}$$

19. **Ans (A)**

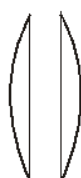


$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{-R} \right) = (\mu - 1) \times \frac{2}{R}$$

$$\frac{1}{f'} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right)$$

$$\frac{1}{f'} = (\mu - 1) \times \frac{1}{R}$$

$$f' = 2f$$



Radius of curvature remain same
so focal length of each part is $2f$.



20. **Ans (B)**

here the maximum voltage across zener diode
can be 80 V if whole current goes through $10 \text{ k}\Omega$
but zener diode conducts when its voltage
is 90 V .

So current through it is zero.

PHYSICS

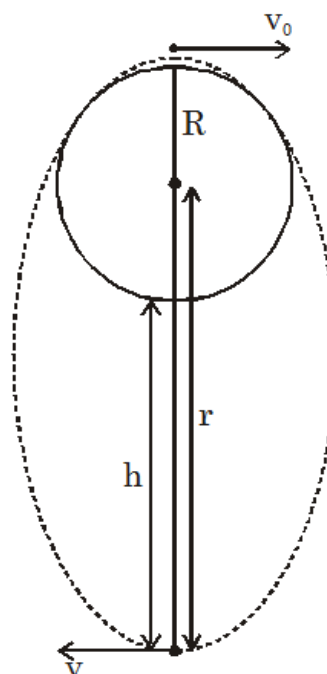
SEC-B

1. **Ans (30.00)**

$$V_A = V_C$$

$$\text{so } C = a + b$$

2. **Ans (0.67)**



$$L = mv_0 R = mvr$$

$$E = \frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

on solving, $h = R$

$$\text{So } \frac{2}{3n} = 1 \Rightarrow n = \frac{2}{3} = 0.67$$

3. Ans (2.82)

Change in kinetic energy = change in potential energy.

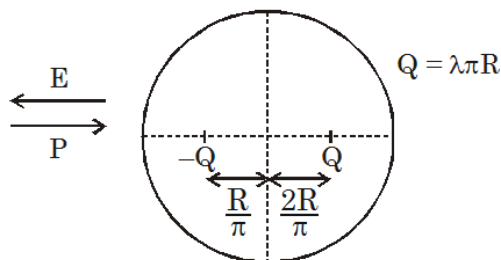
This system can be assumed as dipole of dipole

$$\text{moment } p = \frac{4R}{\pi} \cdot (\lambda\pi R)$$

$$P = 4\lambda R^2$$

$$\text{So } \Delta U = \Delta KE$$

$$PE = \frac{1}{2} I \omega^2$$



$$4\lambda R^2 E_0 = \frac{1}{2} \times m R^2 \omega^2$$

$$\omega = \sqrt{\frac{8\lambda E_0}{m}} = \sqrt{8} \sqrt{\frac{\lambda E_0}{m}}$$

$$n = \sqrt{8} = 2\sqrt{2} = 2 \times 1.41 = 2.82$$

4. Ans (2.00)

$$I = \frac{V}{R} = \frac{200}{80} = 2.5 \text{ A}$$

$$P = \frac{V^2}{R} = \frac{(200)^2}{80} = 50 \text{ W}$$

When the wire is divided in two equal parts and connected in parallel we get maximum power.

Resistance of each part = 40 Ω

$$I_{\max} = 5 \text{ A}$$

$$P_{\max} = \frac{V^2}{R_{\text{eq}}} = \frac{(200)^2}{20} = 2000 \text{ W} = 2 \text{ kW}$$

5. Ans (1.12)

Initially capacitor is charged to $\frac{Q}{2}$.

$$\text{Energy stored in capacitor} = \frac{1}{2} \left(\frac{Q}{2} \right)^2 \times \frac{1}{C}$$

$$= \frac{Q^2}{8C} = \frac{1}{8} CE^2 = U_i$$

Finally energy stored in capacitor,

$$U_f = \frac{1}{2} CE^2$$

$$\text{Work done by the battery} = \frac{3}{2} CE \times E =$$

$$\frac{3}{2} CE^2$$

$$\text{Heat energy lost} = \frac{3}{2} CE^2 - (U_f - U_i)$$

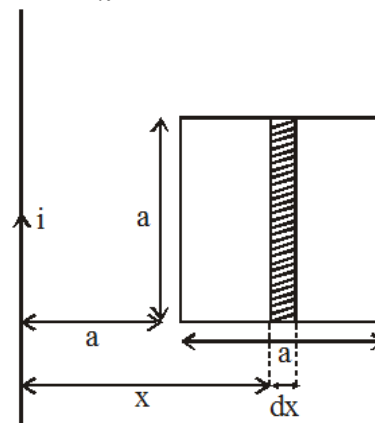
$$= \frac{9}{8} CE^2 = 1.125 CE^2$$

6. Ans (0.34 to 0.35)

$$B = \frac{\mu_0 i}{2\pi x}$$

$$\phi = \int_a^{2a} \frac{\mu_0 i}{2\pi x} dx$$

$$\phi = \frac{\mu_0 i}{2\pi} a \ln 2 = M i$$



$$M = \frac{\mu_0 a \ln 2}{2\pi}$$

$$n = \frac{\ln 2}{2} = \frac{0.693}{2} = 0.35$$

7. Ans (8.00)

KCL of junction :

$$i_1 + i_2 - i_L + i_C = 0$$

$$i_C = i_L - (i_1 + i_2)$$

$$\frac{dq}{dt} = i_L - (i_1 + i_2)$$

$$C \left(\frac{dV_C}{dt} \right) = i_L - (i_1 + i_2)$$

$$2(-6e^{-2t}) = i_L - (10e^{-2t} + 4)$$

$$i_L = -2e^{-2t} + 4$$

$$V_L = -L \frac{di}{dt} = -4 [4e^{-2t} + 0] = -16e^{-2t}$$

$$a = -16, b = 2$$

$$\left| \frac{a}{b} \right| = 8$$

8. Ans (3.21 to 3.24)

$$\text{Required K.E.} = [m(N) + m(n) - m(H) - m(c)]$$

$$\times 931 = 2.99 \text{ MeV}$$

Maximum energy loss

$$= \frac{1}{2} \frac{m_p m_c}{m_p + m_c} v^2 = 2.99 \text{ MeV}$$

$$\Rightarrow \frac{1}{2} m_p v^2 = \frac{14}{13} \times 2.99 \approx 3.228 \text{ MeV}$$

9. Ans (0.50)

$$E_1 = 2\phi, E_2 = 5\phi$$

$$K_{\max} = E - \phi$$

$$\frac{1}{2} m v_1^2 = 2\phi - \phi$$

$$\frac{1}{2} m v_2^2 = 5\phi - \phi$$

$$\frac{v_1^2}{v_2^2} = \frac{1}{4} = 0.25$$

$$\frac{v_1}{v_2} = 0.5$$

10. Ans (2.00)

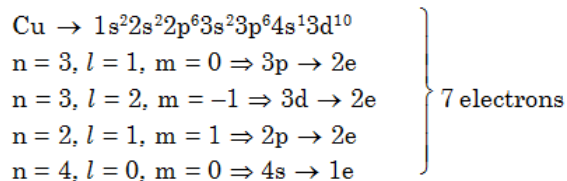
$$\text{Path difference, } (\mu - 1) t = \frac{\lambda}{2}$$

$$t = \frac{\lambda}{2(\mu - 1)} = \frac{4}{2(2 - 1)} = 2 \text{ mm}$$

CHEMISTRY

SEC-A

3. Ans (B)



4. Ans (D)

according to slow step

$$r(t) = K_2 [N_2O_2] [H_2]$$

$$\text{Where } K_1 = \frac{[N_2O_2]}{[NO]^2}$$

$$[N_2O_2] = K_1 [NO]^2$$

$$\text{So, } r(t) = K_2 K_1 [NO]^2 [H_2]$$

$$\text{So, } r(t) = K_2 K_1 [NO]^2 [H_2] K.$$

$$-\frac{1}{2} \frac{d[H_2]}{dt} = -\frac{1}{2} \frac{d[NO]}{dt} = \frac{d[N_2]}{dt} =$$

$$K [NO]^2 [H_2]$$

$$\text{Where } K = K_1 K_2$$

5. Ans (D)

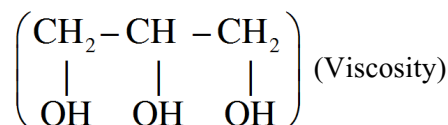
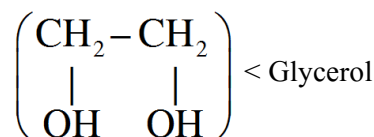
Leaving group ability $\propto$ its stability

Weak base are good leaving group

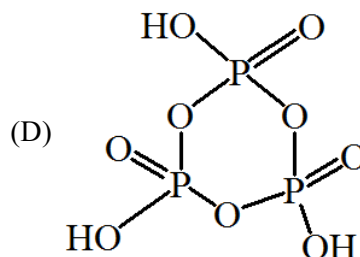
6. Ans (D)

(A) $3p-3p < 3p-3d < 3d-3d$ (π bond strength)

(B) Ethanol (C_2H_5OH) < ethylene glycol



(C) $XeO_3F_2 = XeF_4 = XeF_2$ (All have zero dipole moment)



7. **Ans (C)**

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$6 = 4.74 + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

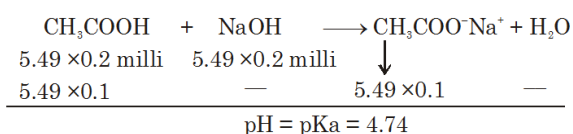
$$1.26 = \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$\frac{[\text{Salt}]}{[\text{Acid}]} = 18.2 = \frac{n \text{CH}_3\text{COO}^- \text{Na}^+}{n \text{CH}_3\text{COOH}}$$

$$18.2 = \frac{100 \times 0.2}{\frac{V \times 0.2}{100 + V}}$$

$$V = 5.49 \text{ ml}$$

Now



10. **Ans (A)**

$$n_{\text{CaCl}_2} = \frac{5.5 \times 10^{-5}}{111} = 5 \times 10^{-5}$$

$$n_{\text{MgCl}_2} = \frac{9.5}{95} = 10^{-4}$$

For hardness in 1 L

$$n_{\text{CaCO}_3} = n_{\text{CaCl}_2} + n_{\text{MgCl}_2}$$

$$n_{\text{CaCO}_3} = 15 \times 10^{-5}$$

$$W_{\text{CaCO}_3} = 15 \times 10^{-3} \text{ gm}$$

$$(\text{Hardness in ppm}) = \frac{15 \times 10^{-3}}{10^3} \times 10^6 =$$

15 ppm

12. **Ans (C)**

(A) Wrought iron is the purest form of iron

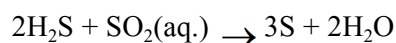
(B) Wolframite is magnetic impurity

(D) Lead and tin are extracted from their chief ore by self reduction and carbon reduction respectively

15. **Ans (D)**

NO_3^- gives brown ring test in presence of conc. H_2SO_4

On passing H_2S gas into aqueous solution of SO_2 , oxidation of H_2S occurs while SO_2 is reduced to elemental sulphur which causes yellow turbidity.



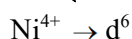
16. **Ans (A)**

Initially the conductance is low due to the feeble ionization of acetic acid. On the addition of base, there is decrease in conductance not only due to the replacement of H^+ by Na^+ but also suppresses the dissociation of acetic acid due to common ion acetate. But very soon, the conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3COOH to CH_3COONa which is the strong electrolyte. This increase in conductance continues raise up to the equivalence point. The graph near the equivalence point is curved due the hydrolysis of salt CH_3COONa . Beyond the equivalence point, conductance increases more rapidly with the addition of NaOH due to the highly conducting OH^- ions.

CHEMISTRY

SEC-B

1. **Ans (60.00)**



Ni is in +4 oxidation state so F^- will act as SFL.

2. Ans (117.00)

T_H = Source Temperature

T_C = Sink Temperature

$$\eta = \frac{T_H - T_C}{T_H}$$

$$\Rightarrow \left(\frac{1}{6} = 1 - \frac{T_C}{T_H} \right) \Rightarrow \frac{T_C}{T_H} = \frac{5}{6}$$

$$\Rightarrow \frac{1}{3} = 1 - \frac{T_C - 65}{T_H} \Rightarrow \frac{T_C - 65}{T_H} = \frac{2}{3} \Rightarrow \frac{5}{6} - \frac{65}{T_H} = \frac{2}{3}$$

$$\Rightarrow \frac{5}{6} - \frac{2}{3} = \frac{65}{T_H} \Rightarrow \frac{5-4}{6} = \frac{65}{T_H} \Rightarrow 65 \times 6 = 390 \text{ K}$$

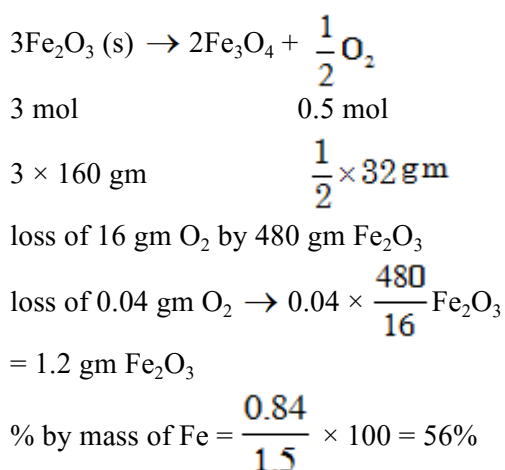
So, Answer is 117°C

4. Ans (5.00)

The mobilities of alkali metal ions in aqueous solutions are : $\text{Li}^+ < \text{Na}^+ < \text{K}^+ < \text{Rb}^+$

BeCl_2 has a polymeric structure in the solid state but a dimeric structure in the vapour state.

5. Ans (56.00)



7. Ans (4.00)

Dielectric constant of D_2O is less than water.

8. Ans (-1656.51)

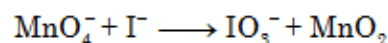
$$\Delta_r H^\circ = \sum \Delta_f H^\circ_{\text{product}} - \sum \Delta_f H^\circ_{\text{reactant}}$$

$$= 3(-110.5) - [3 \times (716.67) - 825]$$

$$= -331.5 - [2150.01 - 825]$$

$$= -331.5 - 1325.01 = -1656.51 \text{ kJ}$$

10. Ans (6.00)



Basic medium is required.

MATHEMATICS

SEC-A

1. Ans (B)

$$\sec^2 x + \csc^2 x - \sin 2x - 3 = 0$$

$$2 + \tan^2 x + \cot^2 x = \sin 2x + 3$$

As $2 + \tan^2 x + \cot^2 x \geq 4$, (using AM $\geq$ GM)

and $\sin 2x + 3 \leq 4$

Here $2 + \tan^2 x + \cot^2 x = 4$ and $\sin 2x + 3 = 4$

$\tan x = \pm 1$ and $\cot x = \pm 1$ and $\sin 2x = 1$

$\tan x = \pm 1$ and $\cot x = \pm 1$ at

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{11\pi}{4}, \frac{13\pi}{4}, \frac{15\pi}{4}$$

and $\sin 2x = 1$ at,

$$2x = \frac{\pi}{2}, 2\pi + \frac{\pi}{2}, 4\pi + \frac{\pi}{2}, 6\pi + \frac{\pi}{2}$$

$$\text{Solutions are } x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$$

Total 4 solutions.

2. Ans (B)

$$\text{Here } s = \frac{6^r}{2^{2r+1} + 3^{2r+1}} = \frac{\frac{6^r}{2^{2r+1}}}{1 + \left(\frac{3}{2}\right)^{2r+1}}$$

$$= \frac{\frac{2^r \cdot 3^r}{2 \cdot 2^{2r}}}{1 + \left(\frac{3}{2}\right)^{r+1} \left(\frac{3}{2}\right)^r},$$

$$= \frac{\frac{1}{2} \left(\frac{3}{2}\right)^r}{1 + \left(\frac{3}{2}\right)^{r+1} \left(\frac{3}{2}\right)^r} = \frac{\left(\frac{3}{2}\right)^r \left(\frac{3}{2} - 1\right)}{1 + \left(\frac{3}{2}\right)^{r+1} \left(\frac{3}{2}\right)^r}$$

$$= \frac{\left(\frac{3}{2}\right)^{r+1} - \left(\frac{3}{2}\right)^r}{1 + \left(\frac{3}{2}\right)^{r+1} \left(\frac{3}{2}\right)^r}$$

$$\begin{aligned} \text{Now } \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{6^r}{2^{2r+1} + 3^{2r+1}} \right) \\ = \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{\left(\frac{3}{2}\right)^{r+1} - \left(\frac{3}{2}\right)^r}{1 + \left(\frac{3}{2}\right)^{r+1} \left(\frac{3}{2}\right)^r} \right) \\ = \sum_{r=1}^{\infty} \tan^{-1} \left(\frac{3}{2} \right)^{r+1} - \tan^{-1} \left(\frac{3}{2} \right)^r \end{aligned}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \tan^{-1} \left(\frac{3}{2} \right)^{r+1} - \tan^{-1} \left(\frac{3}{2} \right)^r$$

$$= \lim_{n \rightarrow \infty} \left(\tan^{-1} \left(\frac{3}{2} \right)^2 - \tan^{-1} \left(\frac{3}{2} \right) \right) + \left(\tan^{-1} \left(\frac{3}{2} \right)^3 - \tan^{-1} \left(\frac{3}{2} \right)^2 \right) \dots$$

$$\dots \left(\tan^{-1} \left(\frac{3}{2} \right)^{n+1} - \tan^{-1} \left(\frac{3}{2} \right)^n \right)$$

$$= \lim_{n \rightarrow \infty} \tan^{-1} \left(\frac{3}{2} \right)^{n+1} - \tan^{-1} \frac{3}{2} = \frac{\pi}{2} - \tan^{-1} \frac{3}{2}$$

$$= \cot^{-1} \frac{3}{2}$$

3. Ans (A)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = k \Rightarrow$$

$$\frac{a}{\sin 55^\circ} = \frac{b}{\sin 15^\circ} = \frac{c}{\sin 110^\circ} = k$$

$$a = k \sin 55^\circ, b = k \sin 15^\circ, c = k \sin 110^\circ$$

$$c^2 - a^2 = k^2 \sin^2 110^\circ - k^2 \sin^2 55^\circ$$

$$= k^2 (\sin(110^\circ + 55^\circ)) (\sin(110^\circ - 55^\circ))$$

$$= k^2 \sin 165^\circ \cdot \sin 55^\circ$$

$$= k^2 \sin 15^\circ \cdot \sin 55^\circ$$

$$= (k \sin 15^\circ) (k \sin 55^\circ) = ab$$

4. Ans (C)

Lines are concurrent then

$$\begin{vmatrix} k & 2 & 2 \\ 2 & k & 3 \\ 3 & 3 & k \end{vmatrix} = 0$$

$$k(k^2 - 9) - 2(2k - 9) + 2(6 - 3k) = 0$$

$$k^3 - 9k - 4k + 18 + 12 - 6k = 0$$

$$k^3 - 19k + 30 = 0$$

$$(k - 2)(k - 3)(k + 5) = 0 \Rightarrow k = 2, 3, -5$$

But for $k = 2$, two lines are parallel for $k = 3$,
 -5 line are concurrent, so only two values.

5. Ans (A)

$$|a| > 1 \Rightarrow 0 < \frac{1}{|a|} < 1$$

$$\text{Now } \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} \frac{1}{a^{i+j+k}}$$

$$= \left(\sum_{i=1}^{\infty} \frac{1}{a^i} \right) \left(\sum_{j=1}^{\infty} \frac{1}{a^j} \right) \left(\sum_{k=1}^{\infty} \frac{1}{a^k} \right)$$

$$= \left(\frac{\frac{1}{a}}{1 - \frac{1}{a}} \right) \left(\frac{\frac{1}{a}}{1 - \frac{1}{a}} \right) \left(\frac{\frac{1}{a}}{1 - \frac{1}{a}} \right) = (a - 1)^{-3}$$

6. Ans (B)

In case of shortest distance normal of the ellipse will also be the normal of the circle

Ellipse $\frac{x^2}{2} + \frac{y^2}{1} = 1$, any point P on the ellipse is $P(\sqrt{2}\cos\theta, \sin\theta)$

Circle $x^2 + y^2 - 3x - 2\sqrt{2}y + 4 = 0$

Centre $\left(\frac{3}{2}, \sqrt{2}\right)$, $r = \sqrt{\frac{9}{4} + 2 - 4} = \frac{1}{2}$

Normal of the ellipse at P $(\sqrt{2}\cos\theta, \sin\theta)$ is $\sqrt{2}x \sec\theta - y \operatorname{cosec}\theta = 1$

This normal will also pass through the centre of the circle $C\left(\frac{3}{2}, \sqrt{2}\right)$

$$\sqrt{2} \times \frac{3}{2} \sec\theta - \sqrt{2} \operatorname{cosec}\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$$

$$P\left(1, \frac{1}{\sqrt{2}}\right) \& C\left(\frac{3}{2}, \sqrt{2}\right)$$

$$\begin{aligned} \text{Minimum distance} &= |PC - r| \\ &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\sqrt{2} - \frac{1}{\sqrt{2}}\right)^2} - \frac{1}{2} \\ &= \sqrt{\frac{1}{4} + 2 + \frac{1}{2} - 2} - \frac{1}{2} = \frac{\sqrt{3} - 1}{2} \end{aligned}$$

7. Ans (B)

Members Partitioning Number of ways

4 Members 1

$$\frac{4!}{1!3!} = 4$$

1 + 3 members

2 + 2 members

$$\left(\frac{4!}{2!2!}\right) \times \frac{1}{2!} = 3$$

1 + 1 + 2 members

$$\left(\frac{4!}{1!1!2!}\right) \times \frac{1}{2!} = 6$$

1 + 1 + 1 + 1 members

$$\left(\frac{4!}{(1!)^4}\right) \times \frac{1}{4!} = 1$$

Total 15 ways

8. Ans (C)

non zero solutions exist when $\Delta = 0$

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

$$R_2 \rightarrow R_2 - R_1 \text{ and } R_3 \rightarrow R_3 - R_1$$

$$\begin{vmatrix} 1 & 2a & a \\ 0 & 3b-2a & b-a \\ 0 & 4c-2a & c-a \end{vmatrix} = 0$$

$$\Rightarrow (3b-2a)(c-a) - (4c-2a)(b-a) = 0$$

$$\Rightarrow 3bc - 3ab - 2ac + 2a^2 - 4bc + 4ac + 2ab - 2a^2 = 0$$

$$\Rightarrow -bc - ab + 2ac = 0 \Rightarrow b = \frac{2a}{a+c}$$

9. Ans (D)

Here we have to find $f'\left(\frac{7}{2}\right)$ so $x \in (3, 4)$,

$$f(x) = [x]^{(x-[x])} + (x-[x])^{[x]} + \sin(\pi x)$$

$$f(x) = (3)^{(x-3)} + (x-3)^3 + \sin(\pi x), \text{ as } x \in (3, 4)$$

$$f'(x) = (3)^{(x-3)} \ln 3 + 3(x-3)^2 + \pi \cos \pi x$$

$$\Rightarrow f'\left(\frac{7}{2}\right) = \sqrt{3} \ln 3 + \frac{3}{4}$$

10. Ans (D)

Tangent of $xy = c^2$ at $\left(ct, \frac{c}{t}\right)$ is

$$\left(\frac{xc}{t} + yct\right) = 2c^2 \Rightarrow x + yt^2 = 2ct \quad \dots(1)$$

$$\text{Slope of tangent} = -\frac{1}{t^2}$$

Equation of line passing (0,0) and

perpendicular to tangent is $y = t^2x \Rightarrow t^2 = \frac{y}{x}$

Now squaring (1), $(x + yt^2)^2 = 4c^2t^2$

for locus of foot of perpendicular replace

$$t^2 = \frac{y}{x}$$

$$\text{Locus is } \left(x + y\left(\frac{y}{x}\right)\right)^2 = 4c^2\left(\frac{y}{x}\right)$$

$$\Rightarrow (x^2 + y^2)^2 = 4c^2yx$$

11. **Ans (A)**

$$\text{Let } K_1 = (x-2)^3(x-3)^2, K_2 = (x-1)^4(x-3)^2, K_3 = (x-1)^4(x-2)^3$$

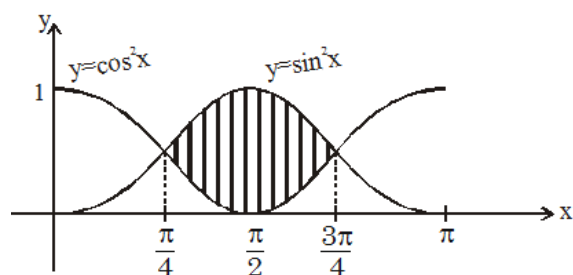
$$\text{Now } f'(x) = K_1 4(x-1)^3 + K_2 3(x-2)^2 + K_3 2(x-3),$$

$$\text{Here } x=3 \text{ is the root of } f'(x)=0. \text{ Hence } f'(3)=0,$$

$$\text{Similarly } x=2 \text{ is the root of } f''(x)=0 \text{ and } x=1 \text{ is the root of } f'''(x)=0, \text{ so } f''(2)=0 \text{ and } f'''(1)=0,$$

$$f'''(1) + 2f''(2) + 3f'(3) = 0$$

12. **Ans (C)**



Area

$$= \int_{\pi/4}^{3\pi/4} (\sin^2 x - \cos^2 x) dx = - \int_{\pi/4}^{3\pi/4} \cos 2x dx$$

$$A = - \left[\frac{\sin 2x}{2} \right]_{\pi/4}^{3\pi/4} = 1$$

13. **Ans (C)**

$$20 \left(\frac{x-2y+1}{\sqrt{5}} \right)^2 + 45 \left(\frac{2x+y+2}{\sqrt{5}} \right)^2 = 180$$

$$\frac{\left(\frac{x-2y+1}{\sqrt{5}} \right)^2}{(3)^2} + \frac{\left(\frac{2x+y+2}{\sqrt{5}} \right)^2}{(2)^2} = 1$$

Lengths of major and minor axes are 6 and 4 respectively.

14. **Ans (C)**

The sum of the coefficients which doesn't contain b is (put $a=c=d=1$ and $b=0$) $= 3^6$

The sum of the coefficients which doesn't contain a and b is (put $a=0, b=0$ and $c=d=1$) $= 2^6$

Sum of the coefficients which contain a but not b is $= 3^6 - 2^6$

15. **Ans (B)**

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{f^2(a+x)}{f(a)} \right)^{\frac{1}{x}} \\ = e^{\lim_{x \rightarrow 0} \left(\frac{f^2(a+x)}{f(a)} \right)^{\frac{1}{x}}} = e^{\lim_{x \rightarrow 0} \left(\frac{f^2(a+x)-f^2(a)}{f(a)} \right)^{\frac{1}{x}}} \\ = e^{\lim_{x \rightarrow 0} \frac{f^2(a+x)-f^2(a)}{x} \times \frac{1}{f(a)}} \quad \text{as } f(a)=1; f^2(a)=1 \\ = e^{\lim_{x \rightarrow 0} \left(\frac{f(a+x)-f(a)}{x} \right) \left(\frac{f(a+x)+f(a)}{f(a)} \right)} \\ = e^{f'(a) \left(\frac{2f(a)}{f(a)} \right)} = e^{2f'(a)} = e^4 \end{aligned}$$

16. **Ans (C)**

According to LMVT

$$|f''(x)| = \left| \frac{f'(x) - f'(0)}{x} \right| \leq 1$$

$$|f'(x)| \leq |x|$$

Again by LMVT

$$\left| \frac{f(x) - f(0)}{x} \right| \leq |x|$$

$$|f(x)| \leq |x|^2$$

Only (C) can be true here.

17. **Ans (C)**

$$\begin{aligned} \int \frac{(3x^2 + 2x) dx}{x^6 + x^4 + 1 + 2x^5 + 2x^3 + 2x^2 + 4} \\ = \int \frac{(3x^2 + 2x) dx}{(x^3 + x^2 + 1)^2 + 2^2}, \quad \text{Put } x^3 + x^2 + 1 = t \\ = \frac{1}{2} \tan^{-1} \frac{(x^3 + x^2 + 1)}{2} + c \end{aligned}$$

18. **Ans (D)**

The plane is

$$(x + 3y + z) + \lambda(zx - y) = 4$$

$$(x + 3y + z - 4) + \lambda(zx - y) = 0$$

This plane always pass through the line of intersection of the planes

$$x + 3y + z - 4 = 0$$

$$2x - y = 0 \Rightarrow \frac{x}{1} = \frac{y}{2} \quad \dots(i)$$

$$\text{Now } x + 3y + z - 4 = 0 \Rightarrow 7x + z - 4 = 0$$

$$\frac{x}{1} = \frac{z-4}{-7} \quad \dots(ii)$$

Plane passes through the line

$$\frac{x}{1} = \frac{y}{2} = \frac{z-4}{-7}$$

19. Ans (B)

$(2,3) \in R$ but $(3,2) \notin R$.

Hence R is not symmetric.

20. Ans (A)

$$p \rightarrow q = \sim p \vee q$$

$$q \rightarrow p = \sim q \vee p$$

$$(p \rightarrow q) \vee (q \rightarrow p) = (\sim p \vee q) \vee (\sim q \vee p)$$

$$\begin{array}{ccccc} p & q & \sim p \vee q & \sim q \vee p & (\sim p \vee q) \vee (\sim q \vee p) \\ T & T & T & T & T \\ T & F & F & T & T \\ F & T & T & F & T \\ F & F & T & T & T \end{array}$$

MATHEMATICS

SEC-B

1. Ans (1.50)

The orthocentre of triangle formed by three tangents of the parabola always lies on directrix of the parabola.

So here locus is directrix

$$(4x-3)^2 = -64(2y+1)$$

$$\Rightarrow \left(x - \frac{3}{4}\right)^2 = -8\left(y + \frac{1}{2}\right)$$

Here equation of directrix is $y + \frac{1}{2} = 2$

$$\Rightarrow y = \frac{3}{2} \Rightarrow \lambda = \frac{3}{2} = 1.5$$

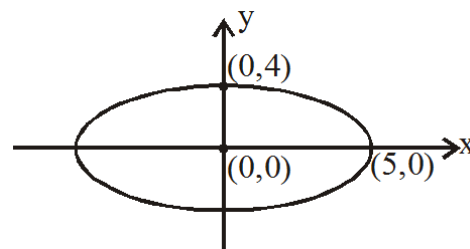
2. Ans (1.00)

$$\text{If } |z - z_1| + |z - z_2| = k$$

If $k > |z_1 - z_2|$ then locus is ellipse where $|z_1 - z_2| = 2ae$ and $2a = k$.

$$\text{here } 2a = 10 \Rightarrow a = 5, 2ae = 6 \Rightarrow e = \frac{3}{5}$$

$$b^2 = a^2 (1 - e^2) \Rightarrow b^2 = 25 \left(1 - \frac{9}{25}\right) \Rightarrow b = 4$$



Here z lies on the ellipse

$$|z|_{\text{maximum}} = |z - (0 + 0i)|_{\text{max}}$$

$$= \text{maximum distance of } z \text{ from } (0,0) = 5$$

$$|z|_{\text{minimum}} = \text{minimum distance of } z \text{ from } (0,0) = 4$$

$$|z|_{\text{max}} - |z|_{\text{min}} = 1$$

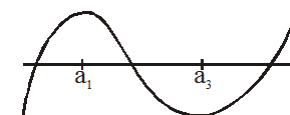
3. Ans (3.00)

$$f(x) = (x - a_1)(x - a_2)(x - a_3) + (a_1 - x) + (a_2 - x) + (a_3 - x), a_1 < a_2 < a_3 \text{ given}$$

as $x \rightarrow \infty, f(x) \rightarrow \infty$ and $x \rightarrow -\infty, f(x) \rightarrow -\infty$

$$f(a_1) = (a_2 - a_1) + (a_3 - a_1) > 0$$

$$f(a_3) = (a_1 - a_3) + (a_2 - a_3) < 0$$



Here $f(x) = 0$ have 3 distinct real roots.

4. Ans (40.00)

$$P\left(\frac{A'}{B}\right)' = 1 - P\left(\frac{A'}{B}\right)$$

$$= 1 - \frac{P(A' \cap B)}{P(B)} = 1 - \frac{P(A \cup B)'}{P(B')}$$

$$= 1 - \left(\frac{1 - P(A \cup B)}{1 - P(B)}\right)$$

$$= 1 - \left(\frac{1 - (P(A) + P(B) - P(A \cap B))}{1 - P(B)}\right) = \frac{2}{15}$$

$$\lambda = \frac{2}{15} \Rightarrow 300\lambda = 300 \times \frac{2}{15} = 40$$

5. Ans (36.00)

$$|\text{adj}(\text{adj}(\text{adj}(\text{adj}A)))| = |A|^{(n-1)^4} = |A|^{16}$$

$$A = \begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow |A| = x + y + z$$

$$\text{given } |A|^{16} = 4^8 \cdot 5^{16} = 2^{16} \cdot 5^{16}$$

$$|A|^{16} = (10)^{16} \Rightarrow |A| = 10$$

$$x + y + z = 10$$

as $x, y, z \in \mathbb{N}$ so positive integral solutions of the given equation are $= {}^{10-1}C_{3-1} = {}^9C_2 = 36$

6. Ans (0.00)

$$f(x) = \begin{cases} x^2 - (x^2 - 1) + 2(-x - 1) - 2x - 7, & x < -1 \\ x^2 + (x^2 - 1) + 2(x + 1) - 2x - 7, & -1 \leq x < 0 \\ x^2 - (1 - x^2) + 2(1 - x) + 2x - 7, & 0 \leq x < 1 \\ x^2 - (x^2 - 1) + 2(x - 1) + 2x - 7, & x \geq 1 \end{cases}$$

$$f(x) = \begin{cases} -4x - 8, & x < -1 \\ 2x^2 - 6, & -1 \leq x < 0 \\ 2x^2 - 6, & 0 \leq x < 1 \\ 4x - 8, & x > 1 \end{cases}$$

$f(x)$ is always continuous

$$f'(x) = \begin{cases} -4, & x < -1 \\ 4x, & -1 \leq x < 0 \\ 4x, & 0 \leq x < 1 \\ 4, & x > 1 \end{cases}$$

$f(x)$ is differentiable for all real x

7. Ans (1.00)

$$f(x) = 1 - x - x^3$$

$$f(x) = -1 - 3x^2 < 0$$

$f(x)$ is decreasing function

$$\text{Now } 1 - f(x) - f^3(x) > f(1 - 5x)$$

$$ff(x) > f(1 - 5x)$$

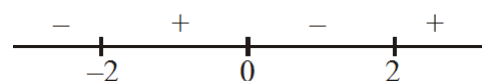
$f(x) < 1 - 5x$, as $f(x)$ is decreasing function

$$1 - x - x^3 < 1 - 5x$$

$$x^3 - 4x > 0$$

$$x(x^2 - 4) > 0 \Rightarrow x(x + 2)(x - 2) > 0$$

$$x \in (-2, 0) \cup (2, \infty)$$



only $x = -1$ satisfying the given inequality.

Number of negative integers are 1.

8. Ans (51.80)

$$S = n \times \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{3n^2 + 8rn + 4r^2}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{\frac{1}{n}}{4\left(\frac{r}{n}\right)^2 + 8\left(\frac{r}{n}\right) \times 3}$$

$$= \int_0^1 \frac{dx}{4x^2 + 8x + 3} = \frac{1}{4} \int_0^1 \frac{dx}{x^2 + 2x + \frac{3}{4}}$$

$$= \frac{1}{4} \int_0^1 \frac{dx}{x^2 + 2x + 1 - \frac{1}{4}} = \frac{1}{4} \int_0^1 \frac{dx}{(x+1)^2 - \left(\frac{1}{2}\right)^2}$$

$$= \frac{1}{4} \times \frac{1}{2 \times \frac{1}{2}} \left[\ln \left(\frac{x+1 - \frac{1}{2}}{x+1 + \frac{1}{2}} \right) \right]_0^1$$

$$= \frac{1}{4} \left[\ln \left(\frac{\frac{3}{2}}{\frac{5}{2}} \right) - \ln \left(\frac{\frac{1}{2}}{\frac{3}{2}} \right) \right]$$

$$= \frac{1}{4} \left[\ln \left(\frac{3}{5} \right) - \ln \left(\frac{1}{3} \right) \right]$$

$$\Rightarrow \lambda = \frac{9}{5} = 1.80$$

9. Ans (37.50)

Let θ_1 is the angle between $\vec{a}$ and $\vec{b}$, θ_2 is the angle between $\vec{c}$ and $\vec{d}$

$$\text{Here } \vec{a} \cdot \vec{b} = \frac{1}{2} = \vec{c} \cdot \vec{d}, \theta_1 = \frac{\pi}{3}, \theta_2 = \frac{\pi}{3}$$

$$k = \left| [\vec{a} \ \vec{b} \ \vec{d}] \vec{c} - [\vec{a} \ \vec{b} \ \vec{c}] \vec{d} \right|$$

$$k = \left| (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) \right|$$

$$k = |\vec{a} \times \vec{b}| |\vec{c} \times \vec{d}| \sin \frac{\pi}{6}$$

$$k = |\vec{a}| |\vec{b}| \sin \theta_1 |\vec{c}| |\vec{d}| \sin \theta_2 \sin \frac{\pi}{6}$$

$$k = \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} \times \frac{1}{2} \Rightarrow k = \frac{3}{8} = .375$$

$$\Rightarrow 100k = 37.5$$

10. Ans (3.20)

$$\frac{dx}{dy} = xy^3 + x^2y^7$$

$$\frac{1}{x^2} \left(\frac{dx}{dy} \right) + \left(-\frac{1}{x} \right) y^3 = y^7$$

$$-\frac{1}{x} = t \Rightarrow \frac{1}{x^2} \frac{dx}{dy} = \frac{dt}{dy}$$

$$\frac{dt}{dy} + ty^3 = y^7, P = y^3, Q = y^7$$

$$IF = e^{\int p dy} = e^{\int y^3 dy} = e^{\frac{y^4}{4}}$$

Solution of differential equation is

$$t \times IF = \int (Q \times IF) dy$$

$$t \times e^{\frac{y^4}{4}} = \int y^7 \cdot e^{\frac{y^4}{4}} dy,$$

$$\frac{y^4}{4} = p \Rightarrow \frac{4y^3}{4} dy = dp$$

$$t \times e^{\frac{y^4}{4}} = \int 4pe^p dp$$

$$t \times e^{\frac{y^4}{4}} = 4(pe^p - e^p) + c \Rightarrow$$

$$-\frac{e^{\frac{y^4}{4}}}{x} = y^4 e^{\frac{y^4}{4}} - 4e^{\frac{y^4}{4}} + c$$

$$\text{as } x = \frac{1}{4}, y = 1$$

$$-4e^{\frac{1}{4}} = e^{\frac{1}{4}} - 4e^{\frac{1}{4}} + c \Rightarrow c = -e^{\frac{1}{4}}$$

$$\frac{-e^{\frac{y^4}{4}}}{x} = y^4 \cdot e^{\frac{y^4}{4}} - 4e^{\frac{y^4}{4}} - e^{\frac{1}{4}}$$

$$\text{at } y = -1$$

$$-\frac{e^{\frac{1}{4}}}{x} = e^{\frac{1}{4}} - 4e^{\frac{1}{4}} - e^{\frac{1}{4}} \Rightarrow x = \frac{1}{4}$$

$$\frac{dy}{dx} = \frac{1}{xy^3 + x^2y^7} \Rightarrow \frac{dy}{dx} \left(\frac{1}{4}, -1 \right) = \frac{1}{-\frac{1}{4} - \frac{1}{16}}$$

$$\left| \frac{dy}{dx} \right| = \frac{16}{5}$$